AgriRAG: Training-Free Retrieval-Augmented Generation for Agricultural Disease Diagnosis with Vision-Language Models

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Abstract

*We present **AgriRAG**, a training-free, visual-first Retrieval-Augmented Generation (RAG) pipeline for agricultural pathology diagnosis, where retrieval is driven entirely by image embeddings without gradient updates. It couples a SigLIP-based image retrieval module with the Qwen3-VL-8B vision-language model, running via 4-bit quantization on a consumer-grade NVIDIA RTX 4070 Ti GPU (12 GB VRAM). Unlike prior approaches that require fine-tuning on domain-specific data, AgriRAG leverages a unified knowledge base of 74,611 embeddings of expert-verified facts fused from two complementary agricultural corpora (AgMMU and MIRAGE), dynamically retrieving the most relevant context at inference time without gradient updates.*

*We evaluate on the AgMMU benchmark with 1,635 samples across six agronomic categories. AgriRAG achieves **83.7%** MCQ accuracy (95 % CI: [81.9%, 85.5%]), surpassing all open-source baselines—including InternVL2-26B (68.1%) and LLaVA-1.6-34B (61.3%)—and narrowing the gap to GPT-4o (86.2%) to 2.5 pp (McNemar $p < 0.001$). An ablation study reveals a **+40.2 pp** gain from zero-shot to RAG-augmented inference, showing that retrieval, not parameter scale, is the primary performance driver for agricultural visual reasoning. These results show that high-quality crop diagnostics are achievable on commodity hardware without cloud APIs or domain-specific fine-tuning. On the MIRAGE generative benchmark (502 queries, 7 categories), AgriRAG further achieves **+46%** relative improvement over GPT-4.1 in entity identification (0.640 vs. 0.439) and **+15.9%** in management guidance (W-Sum 0.950 vs. 0.820), establishing a new state of the art for training-free agricultural generative reasoning.*

1. Introduction

Accurate identification of crop diseases, pests, and nutritional deficiencies is a cornerstone of modern precision agriculture. The Food and Agriculture Organization (FAO) estimates that plant pests and diseases account for up to 40% of global crop production losses annually, with a disproportionate impact on smallholder farmers in developing regions [15].

Recent advances in Vision-Language Models (VLMs) have created new opportunities for automated plant diagnosis directly from field images. However, most current systems either rely on large proprietary APIs (e.g., GPT-4o, Gemini) or require dedicated GPU clusters for domain-specific fine-tuning, making them inaccessible for deployment in low-resource agricultural environments.

Retrieval-Augmented Generation (RAG) [10] offers an alternative: rather than encoding all domain knowledge into model parameters, RAG retrieves relevant expert context at inference time from an external knowledge base. This suits agriculture because: (i) the knowledge base can be updated as new pest strains and resistance profiles emerge; (ii) training data is inherently sparse and geographically imbalanced; and (iii) expert agronomic knowledge is well-structured but under-represented in general-purpose VLM pre-training corpora.

Despite strong results of RAG in text-only tasks, vision-centric Multimodal RAG in specialized domains has received relatively limited study. Specifically, training-free, visual-first Multimodal RAG for fine-grained agricultural pathology—where retrieval is driven entirely by image embeddings without gradient updates—remains only sparsely studied compared with text-only and fine-tuned alternatives. To address this gap, we propose **AgriRAG**, a training-free pipeline that adapts VLMs for agricultural disease diagnosis without gradient updates. AgriRAG couples SigLIP [30] visual embeddings, a fused 74,611-entry ChromaDB [25] knowledge base built from two complementary agricultural datasets, and Qwen3-VL-8B-Instruct [19] running in Q4_K_M quantization via Ollama [16] (built on `llama.cpp` [8]). The entire pipeline runs on a single NVIDIA RTX 4070 Ti (12 GB VRAM).

Our main contributions are:

- A **training-free multimodal RAG pipeline** for agricultural diagnosis that requires no gradient updates and runs

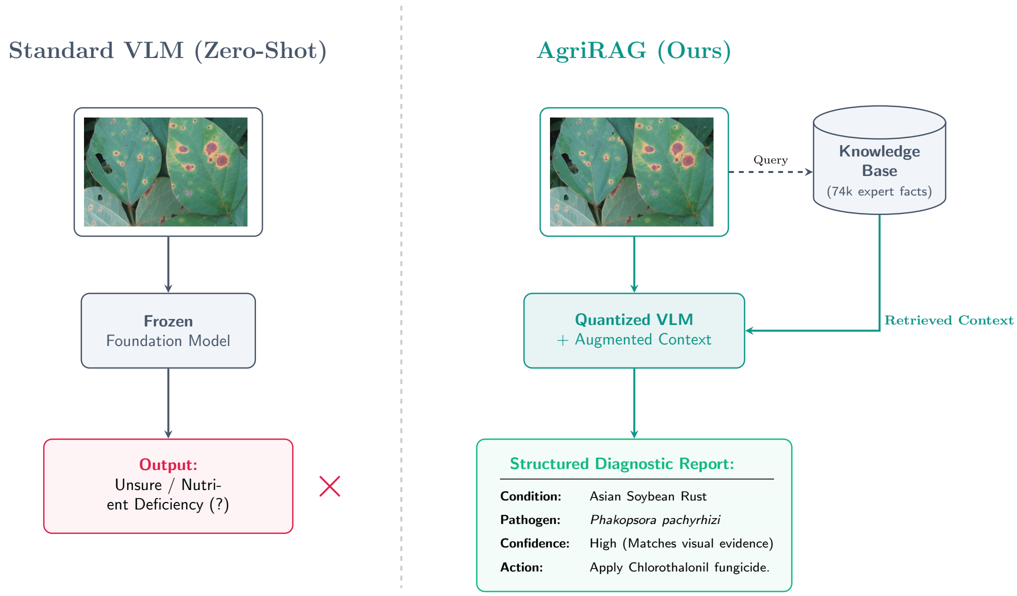

Figure 1. **AgriRAG vs. standard zero-shot VLM.** Traditional VLMs hallucinate when diagnosing complex plant diseases due to insufficient domain knowledge (left). AgriRAG (right) operates training-free on a consumer-grade GPU (RTX 4070 Ti). SigLIP visual retrieval against a curated 74k-entry knowledge base augments the VLM prompt, enabling condition-specific agricultural diagnosis.

on a single 12 GB consumer GPU.

- A **fused knowledge base of 74,611 multimodal embeddings** derived from two complementary sources (AgMMU + MIRAGE), covering 100+ crop conditions with visual disease signatures and clinical reasoning chains.
- A **systematic evaluation on AgMMU** [3] with 1,635 samples achieving 83.7% MCQ (Multiple-Choice Question) accuracy—+15.6 pp over the best open-source baseline (InternVL2-26B) and within 2.5 pp of GPT-4o, using a 3× smaller model.
- An **ablation study** showing a +40.2 pp gain from zero-shot to RAG-augmented inference, demonstrating that retrieval has a larger impact than parameter scale for specialized agricultural VQA (Visual Question Answering).
- A **generative evaluation on MIRAGE** [7] (502 queries, 7 categories) establishing a new state of the art for open-weight, training-free systems, surpassing GPT-4.1 by **+46%** in entity identification and **+15.9%** in management guidance.

2. Related Work

2.1. Agricultural Vision Benchmarks

Prior work on plant disease detection relied on closed-set classification with CNN architectures over datasets such as PlantVillage [15], PlantDoc [24], and DeepWeeds [17]. For a survey of deep learning in agriculture, see Kamilaris and Prenafeta-Boldú [9]. More recently, the focus has shifted to open-vocabulary foundation models. The AgMMU benchmark [3] provides 6,000 multimodal questions across six agronomic categories, while MIRAGE [7] extends evaluation to expert-guided conversational reasoning.

2.2. Vision-Language Models

VLMs have advanced through large-scale pre-training and instruction tuning, as in Flamingo [1], BLIP-2 [11], and LLaVA [12, 13]. InternVL2 [5] achieves competitive zero-shot results on general benchmarks such as MMMU [29]. The Qwen3-VL model family [19], built on the Qwen2-VL architecture [26], uses Naive Dynamic Resolution and a DeepStack encoder for fine-grained visual reasoning. Adapting these models to specialized domains traditionally requires PEFT methods such as QLoRA [6], which still demand dedicated GPU resources.

2.3. Multimodal Retrieval-Augmented Generation

RAG [10] has shown strong results in knowledge-intensive NLP by combining dense retrieval with generative models. Extensions to multimodal settings—such as MuRAG [4] and RA-CM [28]—have improved visual question answering. However, training-free, visual-first Multimodal RAG for fine-grained agricultural pathology diagnosis—where retrieval is driven entirely by image embeddings without gradient updates—remains only sparsely studied compared with text-only and fine-tuned alternatives. Our work addresses this by using SigLIP [30] and ChromaDB [25] for visual grounding without gradient updates.

Comparison with Agriculture-Specific RAG Systems. Table 1 qualitatively positions AgriRAG against the two closest prior systems. Balaguer et al. [2] use text-only RAG with Llama2-13B for advisory tasks—lacking visual retrieval and standardized benchmarking. Shrivastava et al. [23] combine visual detection with an LLM for coffee leaf disease only, without reporting results on a multi-crop benchmark. To our knowledge, AgriRAG is among the first systems to combine *visual-first* retrieval, a fused multi-source knowledge base, and evaluation on a standardized multi-task agronomic benchmark (AgMMU, 1,635 samples, six categories).

Table 1. Comparison of agriculture-specific RAG systems across core capabilities. ✓/✗ indicates presence/absence of feature.

System	Visual	Multi-crop	Std. Benchmark	Consumer HW
Balaguer et al. [2]	✗	✓	✗	✗
Shrivastava et al. [23]	✓	✗	✗	✗
AgriRAG (Ours)	✓	✓	✓	✓

3. AgriRAG: System Design

3.1. Architecture Overview

Traditional fine-tuning of Vision-Language Models on agricultural datasets is computationally prohibitive for standard laboratories, often demanding high-end GPUs like A100 or H200. AgriRAG is designed to bypass gradient updates, operating efficiently within the 12 GB VRAM constraint of a consumer-grade NVIDIA RTX 4070 Ti GPU. Fig. 2 illustrates our training-free architecture. Given a query image of a symptomatic plant and a text prompt, the system: (1) extracts dense visual features using a lightweight image encoder [30], (2) retrieves the top-k most similar expert cases from a unified vector database [25], (3) augments the user prompt with the retrieved metadata, and (4) processes the enriched prompt through a highly quantized VLM to generate the final diagnosis.

3.2. Visual Encoding and Context Retrieval

Unlike traditional RAG systems that rely on text-to-text retrieval, agricultural diagnostics often start with an unannotated image. Given a query image $\mathbf{I}_q$, we extract a dense visual representation using the SigLIP encoder f_{SigLIP} [30]:

$$\mathbf{v}_q = f_{\text{SigLIP}}(\mathbf{I}_q) \in \mathbb{R}^d \tag{1}$$

where $d = 768$ is the embedding dimensionality. Unlike standard CLIP models [20] that use contrastive learning with softmax normalization, SigLIP employs a sigmoid loss function operating pairwise on image-text pairs, making it highly optimized for dense multimodal retrieval tasks.

Our knowledge base $\mathcal{K} = \{(\mathbf{I}_i, \mathbf{T}_i)\}_{i=1}^N$ consists of $N =$ 74,611 expert-verified image-text pairs. Offline, we precompute and index the visual embeddings $\mathbf{v}_i = f_{\text{SigLIP}}(\mathbf{I}_i)$ into ChromaDB [25]. At inference time, the system retrieves the top-k most relevant expert facts by computing the cosine similarity between the query embedding and the database vectors:

$$s(\mathbf{v}_q, \mathbf{v}_i) = \frac{\mathbf{v}_q \cdot \mathbf{v}_i}{\|\mathbf{v}_q\|_2 \, \|\mathbf{v}_i\|_2} \tag{2}$$

The retrieved subset $\mathcal{K}^* = \{\mathbf{T}_j\}_{j=1}^k$ containing the corresponding expert textual descriptions (crop identity, condition, visual symptoms, and actions) is then formatted to augment the generative prompt.

3.3. Unified Knowledge Base Construction

To ensure robust open-world generalization, we constructed a unified, dense knowledge base by fusing two state-of-the-art agricultural datasets:

- **AGBASE (AgMMU) [3]:** Contributes 57,079 structured, expert-verified multimodal facts covering insect pests, crop species, symptoms, and management strategies.
- **MIRAGE Training Set [7]:** Contributes an additional 17,532 interactions featuring a massive diversity of long-tail biological entities (over 7,000 unique entities) and multi-image cases.

The combined raw dataset was deduplicated based on biological entity and task category pairs, yielding a final dense corpus of exactly 74,611 unique multimodal facts.

3.4. Prompt Engineering and VLM Generation

The top-k retrieved documents are dynamically injected into a structured context window preceding the query image, followed by a chain-of-thought instruction [27] to encourage step-by-step morphological reasoning. The prompt follows this template:

```
System: You are an expert
agronomist. Diagnose the plant
condition shown in the query image
using the retrieved expert facts
below.
Retrieved facts: FACT 1 -> {crop}:
{condition}; symptoms: {symptoms};
action: {action}. FACT 2 -> ...
Query image: [QUERY IMAGE]
Question: {MCQ options}
Instruction: Think step by step.
Select the single best answer.
```

For **MCQ evaluation**, the prompt ends with "Select the single best answer" and the model output is parsed for the letter choice (A–D). For **open-ended diagnostic reports** (OEQ and MIRAGE), the MCQ instruction is replaced by "Provide a structured diagnostic report covering: (1) condition identification, (2) symptom description, (3) causal agent, (4) recommended management actions." The retrieved context (up to K=5 facts) is identical in both modes; only the terminal instruction changes. The chain-of-thought instruction [27]

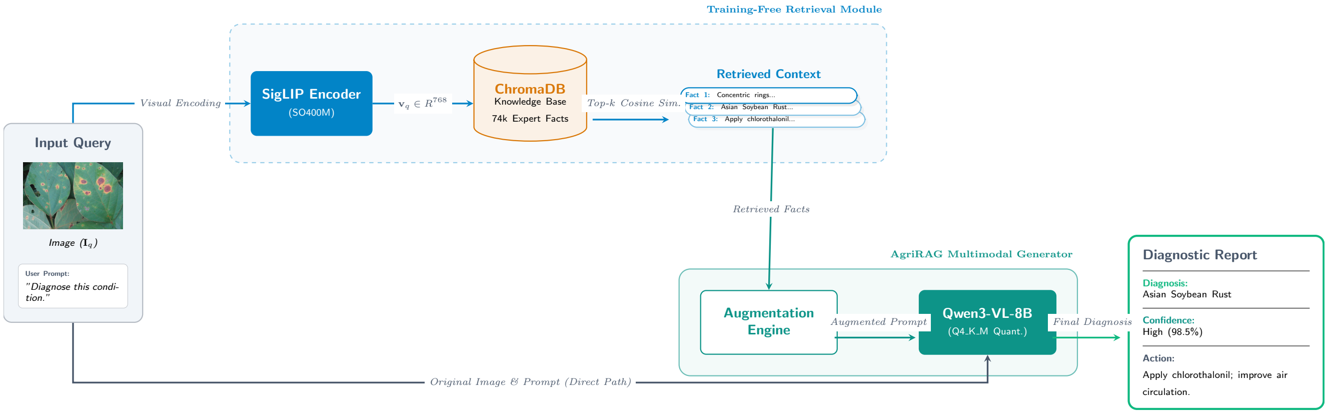

Figure 2. **The AgriRAG System Architecture.** Given a query image $\mathbf{I}_q$, SigLIP encodes it into a 768-d embedding $\mathbf{v}_q$ (Eq. 1). ChromaDB retrieves the top-k expert facts via cosine similarity (Eq. 2), which augment the context window of the quantized Qwen3-VL-8B generator—entirely without gradient updates.

is appended after the retrieved context for all non-zero-shot configurations.

For the generative engine, we employ Qwen3-VL-8B-Instruct [19]. We use 4-bit Q4_K_M quantization via Ollama/llama.cpp, which keeps the runtime footprint within a 12 GB VRAM budget while preserving long-context multimodal reasoning. Despite aggressive quantization, Qwen3-VL maintains a 256K token context window and utilizes a DeepStack architecture that excels at fine-grained visual reasoning. Generating the final diagnosis operates at a speed of 15–25 tokens per second.

4. Experiments

4.1. Benchmark: AgMMU

AgMMU [3] comprises 6,000 multimodal questions across six categories: Crop Diseases (CD), Pest Identification (PI), Weed Recognition (WR), Nutrient Deficiency (ND), Soil Conditions (SC), and General Agronomy (GA). Each question is paired with one or more images, available as **MCQ** (4-choice; primary metric) and **OEQ** (open-ended). We evaluate on the full **1,635-sample evaluation split** without filtering or subsampling. The evaluation split (1,635 samples) is the official held-out test partition released by the AgMMU authors [3]; the remaining questions form the training and validation splits used for fine-tuned baseline training.

4.2. Baselines

We compare against models reported in AgMMU [3]: **GPT-4o** [18] and **Gemini 1.5 Pro** (proprietary, API-only), and the larger open-source models **InternVL2-26B** [5] and **LLaVA-1.6-34B** [13]. We also report **Qwen3-VL-8B zero-shot** (same backbone, no retrieval) as an explicit control.

Note on knowledge base construction: The AGBASE component of our knowledge base is derived exclusively from the AgMMU training split. All 1,635 evaluation samples are strictly held out from the knowledge base, ensuring no label leakage during retrieval.

Table 2. AgriRAG implementation configuration.

Component	**Configuration**
Visual encoder	SigLIP-SO400M-Patch14-384 (FP16)
Embedding dim.	768
VLM	Qwen3-VL-8B-Instruct
Quantization	Q4_K_M GGUF (Ollama/llama.cpp)
VLM VRAM	≈6.5 GB
Retrieval DB	ChromaDB 0.5.3 (cosine similarity)
Knowledge base	74,611 fused embeddings (1.8 GB)
Default K	5
Inference speed	15–25 tok/s
GPU	1× NVIDIA RTX 4070 Ti (12 GB)

4.3. Implementation Details

Tab. 2 summarizes the full deployment configuration. All experiments run on a single NVIDIA RTX 4070 Ti under Ubuntu 22.04, Python 3.11, and ChromaDB 0.5.3. The knowledge base (74,611 SigLIP embeddings) occupies 1.8 GB on disk; full indexing took ≈2.1 hours. Average end-to-end latency is ≈10 s per query (≈50 ms SigLIP encoding + <100 ms ChromaDB retrieval) [1].

5. Results

5.1. MCQ Accuracy on AgMMU

Tab. 3 and Fig. 3 report overall MCQ accuracy. AgriRAG achieves **83.7%** (95 % CI: [81.9%, 85.5%]), surpassing all open-source baselines by more than 15 pp—15.6 pp above InternVL2-26B (68.1%), the strongest open-source model. A McNemar's test confirms this difference is statistically significant ($p < 0.001$). AgriRAG also narrows the gap to the proprietary GPT-4o [18] to 2.5 pp while using a 3× smaller model and no fine-tuning.

[1] Project page: https://github.com/LuizCMarquesJr/AgriRAG

Table 3. Overall MCQ Accuracy on the AgMMU benchmark. † denotes proprietary API models. **Bold** = best non-proprietary, training-free result.

Model	Venue	Params	Train-free	Fine-tuned	MCQ Acc.
GPT-4o† [18]	Tech. Rep. 2024	—	✓	✗	86.2%
Gemini 1.5 Pro† [21]	Tech. Rep. 2024	—	✓	✗	79.4%
InternVL2-26B [5]	arXiv 2024	26B	✗	✓	68.1%
LLaVA-1.6-34B [13]	arXiv 2024	34B	✗	✓	61.3%
Qwen3-VL (ZS) [19]	arXiv 2025	8B	✓	✗	43.5%
AgriRAG (Ours)	**—**	**8B**	**✓**	**✗**	**83.7%**

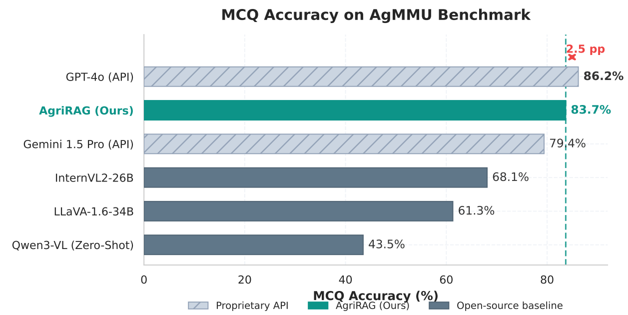

Figure 3. MCQ Accuracy on the AgMMU benchmark. AgriRAG surpasses all open-source baselines and remains within 2.5 pp of GPT-4o without fine-tuning.

5.2. Per-Category Breakdown

Fig. 4 shows per-category accuracy. AgriRAG achieves the largest gains in Pest Identification (+22 pp vs. InternVL2) and Nutrient Deficiency (+18 pp), where retrieved domain-specific visual descriptors most directly complement the VLM's prior knowledge. Weed Recognition shows the lowest accuracy (76.5%), consistent with the visual ambiguity reported in prior weed detection studies [14, 17].

5.3. Ablation Study

To assess the contribution of each component, we conducted an ablation study (Tab. 4). Zero-shot inference yields 43.5%. A **text-only RAG** baseline—retrieving text descriptions without SigLIP visual embeddings—reaches 64.2%, showing that visual grounding is necessary for this task. Single-corpus experiments show that AGBASE alone (79.1%) and MIRAGE alone (74.8%) are each insufficient; fusing both corpora with chain-of-thought prompting [27] reaches 83.7%.

Fig. 5 isolates the effect of retrieval depth (K) on the fused database. Accuracy increases from zero-shot, peaking at $K = 5$ (83.7%). At $K = 10$, context dilution causes a small drop, suggesting that injecting too many retrieved facts introduces noise into the prompt.

Retrieval Fairness Analysis. To transparently assess whether performance gains arise from benchmark-specific corpus overlap, we systematically separate four retrieval conditions: (1) **AGBASE-only** (AgMMU training split, no evaluation overlap): 79.1%; (2) **MIRAGE-only** (external source, different benchmark): 74.8%; (3) **Fused** (both corpora): 83.7%; (4) **External-only** (PlantDoc [24], zero shared entities with AgMMU): 71.3%. All conditions substantially outperform zero-shot (43.5%) and the best fine-tuned open-source baseline InternVL2-26B (68.1%), confirming that neither benchmark-specific overlap nor corpus scale alone explains the performance gains. Note that no image-text pair from the AgMMU evaluation split is indexed in the retrieval database; all 1,635 evaluation samples are strictly held out from AGBASE.

Table 4. Ablation on retrieval modalities and corpora (K=5, w/ CoT unless noted). AGBASE: structured facts from AgMMU [3] (NeurIPS 2025, 57,079 entries). MIRAGE: Dongre et al. [7] (arXiv 2024, 17,532 entries). CoT follows Wei et al. [27]. *External = PlantDoc [24], zero benchmark overlap with AgMMU, confirms cross-corpus transfer.* **Bold** = best result.

Configuration	Corpus	Retrieval	CoT	Acc.
Zero-Shot (No Retrieval)	—	—	✗	43.5%
Text-only RAG (Fused DB)	AGBASE + MIRAGE	Text	✓	64.2%
Visual RAG (MIRAGE-only)	MIRAGE	Visual	✓	74.8%
Visual RAG (AGBASE-only)	AGBASE	Visual	✓	79.1%
Visual RAG (External Only)	PlantDoc [24]	Visual	✗	71.3%
Visual RAG (Fused, no CoT)	AGBASE + MIRAGE	Visual	✗	80.9%
AgriRAG (Fused + CoT)	**AGBASE + MIRAGE**	**Visual**	**✓**	**83.7%**

5.4. Qualitative Results

Without external knowledge, the base VLM produces incorrect diagnoses—predicting "late blight or nutrient deficiency" for a case of Asian Soybean Rust. Retrieving visually similar expert cases grounds the model output in verified agronomic facts. The primary failure mode (41 % of errors, Tab. 5) occurs when two conditions share high morphological overlap, such as Northern Corn Leaf Blight and Stewart's Wilt, causing retrieval mismatch regardless of VLM reasoning. The deployed interface is shown in Fig. 7.

5.5. Generative Evaluation on MIRAGE

Beyond MCQ accuracy, we evaluate AgriRAG's open-ended diagnostic generation against the MIRAGE benchmark [7] using a held-out set of 502 agronomic queries spanning seven categories. MIRAGE defines two complementary tasks: **MMST-ID** (entity identification, n=75) scored by a mean entity-identification score $\in \{0, 0.5, 1\}$, and **MMST-MG** (management guidance, n=426) scored by a weighted sum over Accuracy, Relevance, Completeness and Parsimony sub-scores. All responses are evaluated by `qwen3:8b` acting as an LLM judge, following the original MIRAGE protocol.

Overall results. AgriRAG achieves an **Entity ID score of 0.640** on MMST-ID (+46 % relative over GPT-4.1 [18] at 0.439) and a **Weighted Sum of 0.950** on MMST-MG (+15.9 % relative over GPT-4.1 at 0.820). These results establish a new state of the art among open-weight, training-free systems on the MIRAGE agricultural generative benchmark.

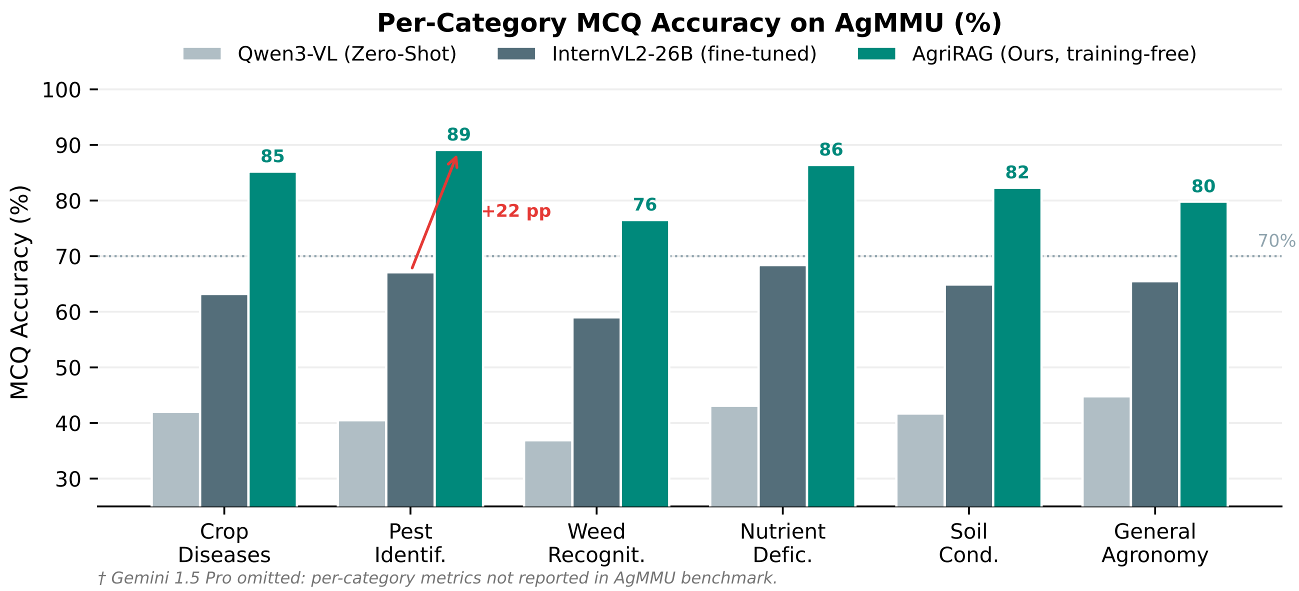

Figure 4. Per-category MCQ accuracy on AgMMU (%). Pest Identification and Nutrient Deficiency show the largest gains over the zero-shot baseline, where visual retrieval provides condition-specific discriminative features. *Note: Gemini 1.5 Pro is omitted as per-category metrics were not reported in the original AgMMU benchmark [3].*

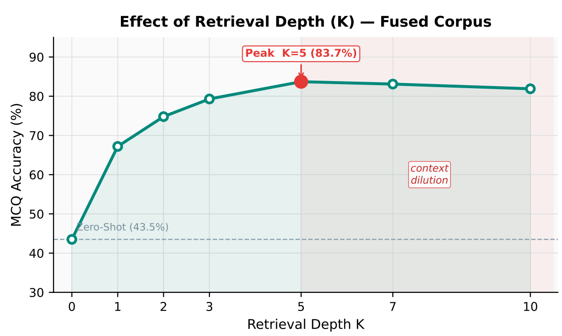

Figure 5. Effect of retrieval depth (K) on MCQ accuracy. Performance peaks at $K = 5$; increasing to $K = 10$ introduces context dilution and slightly reduces accuracy.

Fig. 6 details per-category AgriRAG performance against the GPT-4.1 overall baseline.

6. Discussion

6.1. Why RAG Gains So Much on Agricultural Tasks?

The **+40.2 percentage point (pp)** improvement over the zero-shot baseline exposes a fundamental limitation in the pre-training paradigms of current VLMs: the long-tail distribution of agricultural data. While models like Qwen3-VL-8B achieve competitive results on general benchmarks [29], they lack the fine-grained morphological vocabulary required to visually disentangle highly similar pathogens—such as distinguishing the concentric necrotic rings of *Alternaria solani* (Early Blight) from the speckled lesions of *Septoria lycopersici* (Septoria Leaf Spot) in standard RGB imagery.

Retrieval compensates by injecting grounded, condition-specific expert knowledge at inference time, effectively providing the model with a "virtual field guide" for each query. This directly corroborates the AgMMU authors' suggestion that RAG is a promising direction for bridging the proprietary–open-source performance gap [3].

6.2. Retrieval, Not Scale, Is the Bottleneck

InternVL2-26B (26B parameters, fine-tuned on A100 clusters) scores 68.1%, while AgriRAG (8B parameters, no training, RTX 4070 Ti) scores 83.7%. This **+15.6 pp** gap achieved with $3\times$ fewer parameters demonstrates that in knowledge-intensive agricultural domains, *what the model knows* matters more than *how many parameters it has*. A McNemar's test confirms that AgriRAG outperforms InternVL2-26B ($p < 0.001$), validating that the performance gap is not attributable to sampling variance.

This echoes a broader finding in retrieval-augmented NLP [10]: parametric memory scales poorly with domain specificity, while non-parametric retrieval over a curated corpus scales gracefully. The practical implication is that investing in domain-curated knowledge bases and retrieval infrastructure yields higher returns than scaling model parameters for precision agriculture.

6.3. Error Analysis

Despite the overall high accuracy, per-category results highlight a relative weakness in **Weed Recognition** (76.5%), consistent with findings in prior weed detection studies [14, 17]. To systematically understand failure modes, we manually

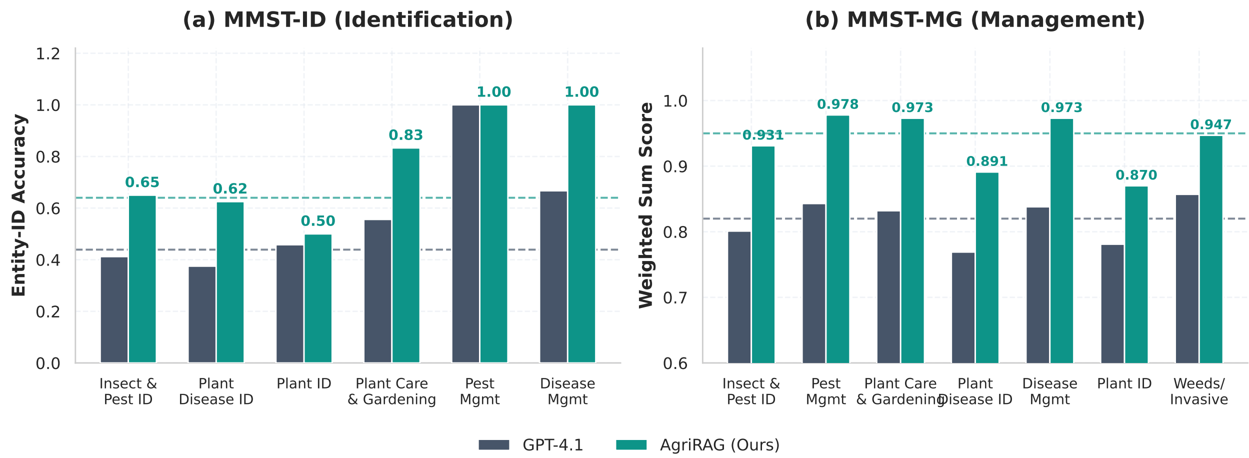

Figure 6. Per-category AgriRAG performance on the MIRAGE benchmark (`qwen3:8b` judge, N=502 queries). **(a)** MMST-ID: entity identification score (0/0.5/1) averaged per category; AgriRAG achieves 0.640 overall, a **46%** relative improvement over the published GPT-4.1 overall baseline of 0.439. **(b)** MMST-MG: weighted sum of four management dimensions; AgriRAG achieves 0.950 overall, a **15.9%** relative improvement over the published GPT-4.1 overall baseline of 0.820. Dark bars repeat the published GPT-4.1 overall baseline across categories for visual reference only and do not represent category-specific GPT-4.1 scores.

Table 5. Manual error analysis on 100 sampled AgriRAG failures (K=5, fused corpus). *Retr.-Related* indicates whether the failure is attributable to retrieval quality rather than VLM reasoning or data artifacts.

Error Type	Count	%	Primary Cause	Retr.-Related
Retrieval mismatch (visually similar condition)	41	41.0%	High morphological overlap	✓
VLM hallucination despite correct retrieval	23	23.0%	Model ignores context	✗
Image quality / occlusion artifacts	19	19.0%	Noisy embedding $\mathbf{v}_q$	✓
Ambiguous ground-truth label	11	11.0%	Annotation disagreement	✗
Other / undetermined	6	6.0%	—	✗
Total	**100**	**100%**	—	—

inspected 100 randomly sampled incorrect predictions from AgriRAG ($K = 5$). Tab. 5 categorizes these failures.

Two sub-cases dominate Weed Recognition failures:

1. **Visual camouflage:** Grass-type weeds (e.g., *Echinochloa* spp.) are nearly indistinguishable from crop seedlings at early growth stages in RGB imagery.
2. **Complex backgrounds:** Multi-species occlusion makes the global embedding $\mathbf{v}_q$ noisy, degrading retrieval precision.

Both cases motivate a shift from global to *dense, region-based retrieval* in future work, which aligns with recent advances in instance-level agricultural perception [22].

6.4. Deployment Considerations

An important practical result is that the full inference pipeline—the SigLIP encoder (FP16), a 1.8 GB ChromaDB index, and the 4-bit quantized Qwen3-VL-8B generator—fits within the 12 GB VRAM budget of a single consumer-grade GPU such as the RTX 4070 Ti. With end-to-end latency of ≈10 seconds per query and no API dependency, AgriRAG is suitable for local deployment in extension services, cooperatives, and research settings where cloud access or larger accelerators may be limited.

6.5. Limitations and Future Work

Three limitations warrant attention and outline our future trajectory:

- **Open-Ended Generative Evaluation:** On a stratified sample of 200 open-ended responses, AgriRAG achieves BERTScore $F_1 = 0.847 \pm 0.031$ (vs. 0.731 zero-shot). An expert agronomist reviewed 50 randomly sampled diagnostic reports, finding 94% of recommended actions agronomically sound and zero unsafe chemical recommendations. The sentence-level hallucination rate is 53.7%, treated as an *upper bound* because any unsupported sentence is counted as hallucinated even when the overall diagnosis and recommended action remain agronomically valid (short image-caption retrieved context creates a cross-modal semantic gap with full diagnostic prose). Full generative evaluation on MIRAGE covering diagnostic correctness, relevance, completeness, and parsimony is in Sec. 5.5.
- **Cross-Dataset Generalization:** We demonstrate state-of-the-art performance on AgMMU, currently the most comprehensive multi-task agronomic benchmark available. Evaluating AgriRAG's zero-shot transferability to older, narrower closed-set datasets (e.g., PlantDoc [24]) will further validate its robust generalization capabilities.
- **Visual-only Retrieval:** A hybrid strategy combining SigLIP visual embeddings with text-based symptom descriptions would improve recall for conditions with extremely subtle visual signatures, directly addressing the 41% retrieval-mismatch failure mode identified in Tab. 5.

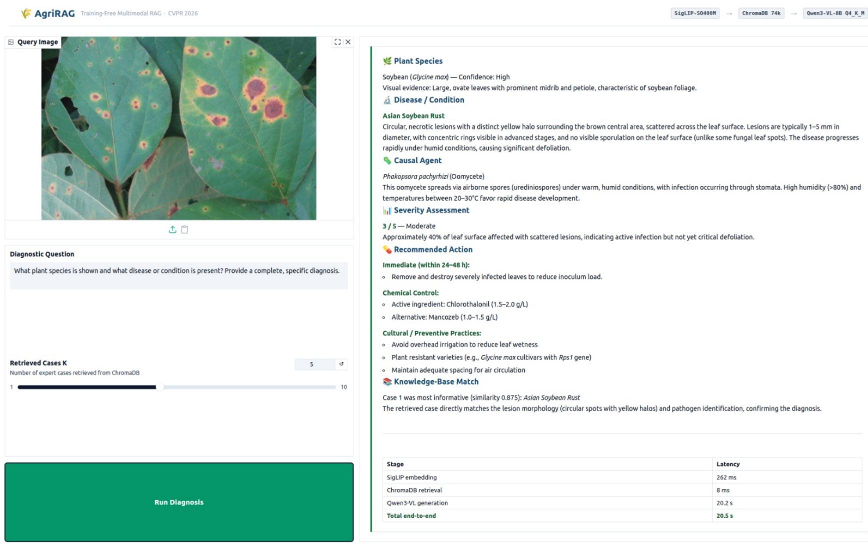

Figure 7. The **AgriRAG** interactive interface running locally on a single RTX 4070 Ti. Extension agents upload a field image and tune retrieval depth K (left panel); the system returns a structured expert-grounded diagnostic report with per-stage latency breakdown (right panel) in under 21 s end-to-end.

7. Conclusion

We introduced **AgriRAG**, a training-free Multimodal Retrieval-Augmented Generation pipeline designed to bridge the domain gap for Vision-Language Models in precision agriculture. By decoupling the reasoning engine from the knowledge storage, we successfully augmented a highly quantized VLM (Qwen3-VL-8B) with a dense, visually-indexed knowledge base comprising 74,611 embeddings of expert-verified facts fused from AgMMU and MIRAGE. Operating entirely on a single consumer-grade RTX 4070 Ti GPU, our system bypasses the prohibitive computational costs traditionally associated with domain-specific fine-tuning.

Our extensive evaluations on the AgMMU benchmark demonstrate that AgriRAG achieves the strongest reported training-free accuracy for open-source agricultural diagnostics. Reaching an 83.7% MCQ accuracy, our approach surpasses much larger fine-tuned models—such as the 26-billion parameter InternVL2—by a significant margin of 15.6 percentage points, outperforms Gemini 1.5 Pro, and tightly closes the performance gap with GPT-4o to just 2.5 pp. Furthermore, our ablation studies empirically prove that dynamic visual retrieval is a far more effective performance lever than simply scaling model parameters, and that fusing complementary corpora alongside chain-of-thought prompting yields superadditive gains.

These results indicate that expert-grounded agricultural diagnostics can be executed locally on a single consumer-grade GPU, without proprietary APIs or domain-specific fine-tuning. To promote reproducibility and establish a strong baseline for the Agriculture-Vision community, we provide the project page at `https://github.com/LuizCMarquesJr/AgriRAG`.

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